CIRCUIT LAB ANSWER SHEET 2018
All answers must be recorded on this sheet

Team name: KEY **Team number:** _____

Members: 1. _____

2. _____

Part I: Written Test

1. B (2)	11. B (2)	21. A (2)
2. A (3)	12. D (2)	22. D (3)
3. B (3)	13. D (2)	23. A (3)
4. B (2)	14. A (2)	24. C (3)
5. D (2)	15. D (2)	25. B (3)
6. E (2)	16. E (2)	26. B (3)
7. A (2)	17. B (2)	27. A (3)
8. C (3)	18. B (2)	28. C (2)
9. D (2)	19. B (4)	29. C (3)
10. A (2)	20. C (2)	

Part II: Hands-On Tasks

Station I

30. (1) C. Capacitor (3)

(2) A. Battery (3)

(3) D. Diode (3)

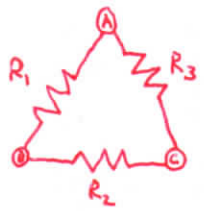
(4) E. Insulator (3)

(5) B. Resistor (3)

Station II

31. Either the "Delta" or "Wye" configuration is acceptable.

Delta:



Where
R_1 between 31.3 kΩ and 34.7 kΩ
R_2 between 53.2 kΩ and 58.8 kΩ
R_3 between 37.0 kΩ and 41.0 kΩ

Wye:



Where
R_1 between 9.1 and 11.0 kΩ
R_2 between 13.3 and 15.6 kΩ
R_3 between 15.8 and 18.4 kΩ

Award 5 points for drawing a correct diagram and 5 points for each correct resistance (20 points total)

Lake Erie / Niagara Invitational - Repaired practice edition - Answer key

Division C | Circuit Lab | Repair appendix

This appendix contains authored replacements and supplied written alternatives. It does not reconstruct unpublished historical material. For the listed item IDs, use this appendix instead of the archived wording and key.

ANSWER KEY - Review only after submitting.

repair-case. (4 points) Authored written substitute for unavailable source material: Why can real charging exceed the ideal time?

Unit: LED matching, charge-time estimates, and troubleshooting.
Worked-case data: Estimate ideal time for 1.5 Ah supplied at constant 0.30 A.

answer: Current tapers and efficiency, temperature, and control constraints matter. (4 points).